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CPU Scheduling Simulator – Comparison Report

Introduction

For this homework, I made a CPU scheduling simulator using Python. The program runs five different scheduling algorithms: FCFS, SJF, Round Robin, Priority, and MLFQ.

I used CSV files to enter the processes and compared the waiting time, turnaround time, and response time for each algorithm.

Scheduling Algorithms

FCFS: Runs processes in the order they arrive. Once a process starts, it finishes before the next one runs.

SJF: Runs the shortest available process first. This can help reduce waiting time.

Round Robin: Each process gets 4 time units. If it doesn't finish, it goes back into the queue.

Priority: Runs processes based on priority. A lower priority number means the process goes first.

MLFQ: Uses three queues with time quantums of 4, 8, and 16. Processes start in the highest queue and move down if they don't finish during their time quantum. I also used a priority boost every 50 time units.

Sample Workload Results

I first tested the simulator using the four processes from the lecture.

Algorithm	Avg Wait	Avg Turnaround	Avg response
FCFS	7.50	12.25	7.50
SJF	4.50	9.25	4.50
Round Robin	8.50	13.25	5

Priority	9	13.75	9
MLFQ	8.50	13.25	5

SJF had the lowest average waiting time at 4.50. Priority had the highest waiting time at 9.00. Round Robin and MLFQ had better response times than FCFS.

Custom Workload Results

I also tested the program with a different workload containing five processes with different arrival times and burst times.

Algorithm	Avg Wait	Avg Turnaround	Avg response
FCFS	8.20	13.80	8.20
SJF	6.00	11.60	6.00
Round Robin	9.60	15.20	5
Priority	6.60	12.20	6.6
MLFQ	10.40	16.00	3.80

For my custom workload, SJF had the lowest waiting and turnaround times again. MLFQ had the lowest response time, but it also had the highest waiting time.

Comparison

After testing both workloads, I noticed that SJF performed the best for waiting and turnaround time. This makes sense because shorter processes get completed first.

Round Robin and MLFQ gave processes CPU time earlier, but some processes had to wait for another turn to finish.

FCFS was the simplest algorithm, but shorter processes could get stuck waiting behind longer ones. Priority Scheduling depended on which processes had higher priority.

Conclusion

This homework helped me understand how CPU scheduling works and how different algorithms affect process performance.

I learned that SJF can reduce waiting time, while Round Robin and MLFQ can improve response time. The results also showed me that an algorithm with a good response time doesn't always have the best turnaround time.